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ABSTRACT

Generic Tail Control Missile (GTCM) models with grid fin and planar fin configurations were studied at the Air Force Research Laboratory Munitions Directorate, Eglin AFB, FL. These tests were conducted in the USAF Aeroballistic Research Facility (ARF). This paper will present the aerodynamic results of these investigations. The tests, which covered a Mach number range of 0.39 to 1.60, indicate increased drag, decreased static stability, and decreased pitch damping for the grid fin configuration. Also, the ballistic range tests identified a critical flow transition at Mach 0.77 where normal shock waves are present within the grid cells resulting in large changes to the center of pressure thereby affecting the static stability.

LIST OF SYMBOLS

A	=	cross section area of the model ($\pi d^2/4$) and reference area
CG	=	center of gravity
$C_{m\alpha}$	=	pitching moment coeff. Derivative
C_{mq}	=	pitch damping moment coefficient
$C_{N\alpha}$	=	normal force coefficient derivative
C_{X_0}	=	zero yaw axial force coefficient
d	=	diameter of the model and reference length
I_x, I_y	=	moment of inertia about the x and y axis
L	=	model length
X	=	downrange position coordinates
X_{CG}	=	location of center of gravity from nose
$X_{c.p.}$	=	location of center of pressure from nose
ψ	=	yaw angle

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INTRODUCTION

A grid fin, also known as lattice control, is an unconventional missile fin comprising an outer frame supporting a grid of lifting surfaces. The purpose of this research effort is to improve the knowledge on the aerodynamics associated with grid fin missile configurations as well as to determine a possible scaling effect for sub-scale model testing.

The data contained in this paper result from two test series conducted on free-flight models at the USAF Aeroballistic Research Facility (ARF).

Background

Grid fins have been studied for several years. This fin design offers favorable lift characteristics at high angle of attack and almost zero hinge moments allowing the use of small and light actuators[1,2,3,4]. In addition they promise good storability for potential tube launched and dispenser launched applications[5]. The typical drawback for grid fins, its high drag, is under investigation. Some promising results have already been made[6].

The available data on grid fins are mainly based on wind tunnel tests and computational fluid dynamic calculations[7,8].

The free-flight ballistic range has several advantages compared to wind tunnels for aerodynamic testing. The most important one is that the test object is in unrestrained flight; therefore, no model support (sting) or wall interference effects are present during the measurement of the data.

Secondly, by applying an initial starting attitude or by disturbing the flight path of the model upon launch the resulting pitch and yaw motion enables the test engineers to determine the dynamic derivatives and coefficients.

One challenge in free flight testing is the model design and construction. Since the model and sabot package is gun launched this package must be sized to fit the launcher and be capable of withstanding the in–gun accelerations. Therefore, the size of the models is often smaller than wind tunnel models and this can result in very high precision requirements during manufacturing. The in-bore acceleration requirements means that the model/sabot design must take into account the high loading during firing. Hence, the models require adequate strength to be launched without suffering structural damage.

In addition to the challenge of building and launching sub-scale grid fin models is the ability to understand the results. It is unclear how the aerodynamics of grid fin configurations will scale. That is, will the results obtained in the free-flight facility be representative of the full-scale flight data?

EXPERIMENTAL FACILITY

The tests were conducted in the Aeroballistic Research Facility (ARF)[9]. This facility is operated and maintained by the Air Force Research Laboratory Munitions Directorate, Eglin AFB, FL. The ARF is an enclosed, instrumented, concrete structure used to examine the exterior ballistics of various free-flight projectiles. The 207-meter instrumented length of the range has a 3.66 m^2 square cross section for the first 69 meters and a 4.88 m^2 square cross section for the remaining length. The range has 131 locations available as instrumented stations of which 50 are currently used to house fully instrumented orthogonal shadowgraph stations. Besides the shadowgraph stations the facility contains one laser-lighted photographic station located in the uprange end of the instrumented section. The range is an atmospheric test facility where the temperature and the relative humidity are controlled to 22 ±1 °C and less than 55% respectively. A chronograph system provides the times for the projectile at each station. These times together with the spatial position and orientation obtained from the orthogonal photographs provide the basic trajectory data from which the aerodynamic coefficients are extracted.

MODEL DESIGN AND TESTING CONDITIONS

Two models were designed and tested for this effort. Both models had exactly the same missile body with only the fins being different. This allows for one-to-one comparison of the data to isolate the fin influence. The design of the models is the result of an international agreement bringing different research activities on lattice controls together. The models themselves featured a three-caliber tangent-ogive nose made from steel mounted to a 13 caliber long cylindrical shaped body made from aluminum. A roll pin was attached to the base of the model in order to be able to collect roll data. The grid fins and planar fins were located on their respective model body such that their aerodynamic center was at a common location.

The grid fin model, shown in Figure 1, incorporates 4 grid fins that have a rectangular shape and were mounted in cruciform orientation. The grid fins were manufactured via electrical discharge machining (EDM) process. This technique was required to construct the thin webs of the grid with the required accuracy. The designed thickness for the grid fin webs was .125 mm. However, it was noted that the thickness of the actual fins varied from .125 mm to .175 mm.

The planar fin model is shown in Figure 2. The clipped delta wings have a double wedge cross-section and were mounted in cruciform orientation. They were installed on the missile body with an angle of incidence of one degree in order to achieve a desired roll rate.

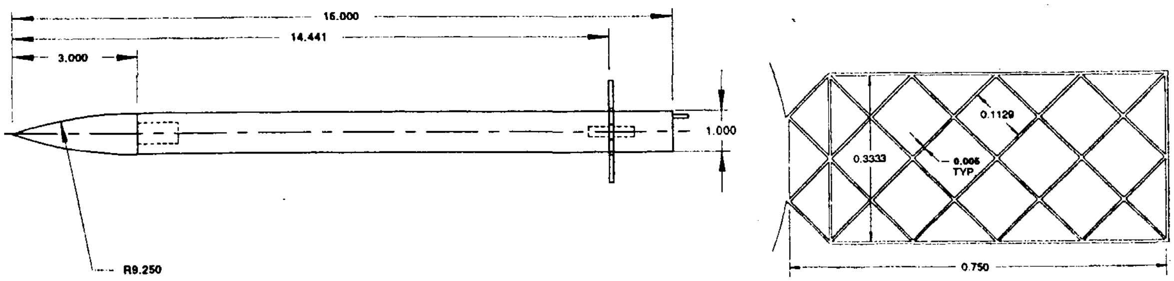

Figure 1 Generic tail control missile (GTCM) with grid fins

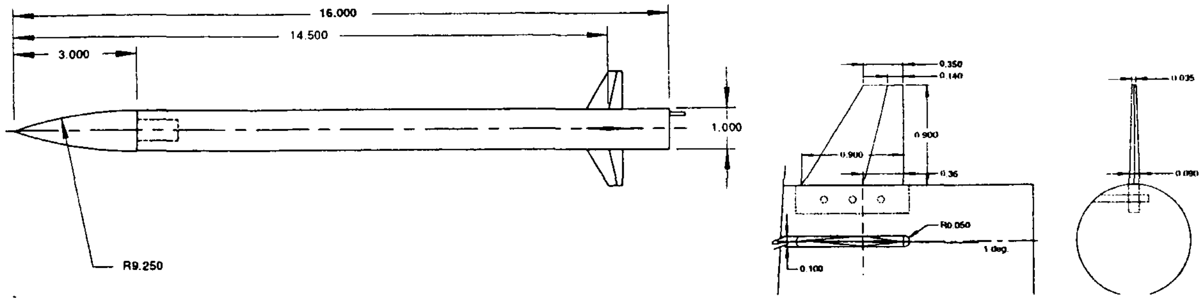

Figure 2 Generic tail control missile (GTCM) with planar fins

Table 1 lists the nominal physical properties for the two Generic Tail Control Missile (GTCM) model configurations. In total 18 GTCM models were manufactured and tested (14 with grid fins and 4 with planar fin). They were fired using a 79 mm high performance gun. One model was lost due to a launch failure. The initial muzzle velocities ranged from 133 m/sec to 566 m/sec.

Table 1: Model physical properties

Configuration	GTCM w/ grid fins	GTCM w/ planar fins
Diameter (d), mm	25.4	25.4
Length (L), mm	406.4	406.4
Mass, g	654.3	646.0
I_x, g cm2	504.	501.
I_y, g cm2	91,624.	89,246.
X_{CG}, mm from nose	193.6	191.8

RESULTS

Flow Field

A recent paper[10] presented the flow field data for the tests documented herein. In this paper were shown the complex shock structure about the current configuration as well as a development of the shock waves through the subsonic, transonic, and supersonic regimes.

Aerodynamic Parameter Identification

From each set of free flight motion data the aerodynamic force and moment coefficients have been extracted. The essential steps of the data reduction are: (1) assemble the physical properties and atmospheric conditions, (2) film reading to determine the spatial position and orientation of the model, (3) mathematically modeling the test configuration's theoretical equation of motion, and (4) matching these theoretical equations to the experimental data in order to determine the aerodynamic parameters of the model. The complete process is described in Reference 11. The Comprehensive Automated Data Reduction and Analysis System (CADRA)[12] is used to read the film and calculate the trajectory. The trajectory matching process is accomplished using the Aeroballistic Research Facility Data Analysis System (ARFDAS)[11,13] and is depicted graphically in Figure 3.

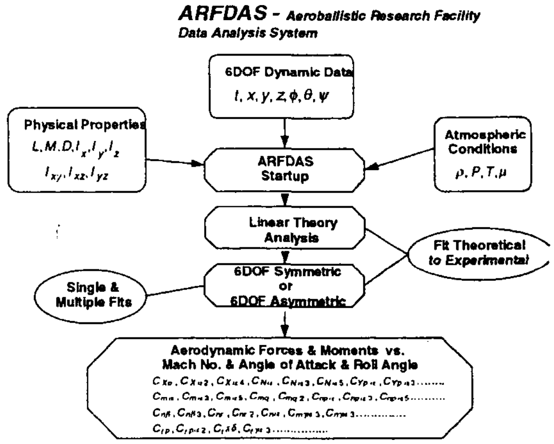

Figure 3 ARFDAS parameter identification process

ARFDAS incorporates a standard linear theory analysis[11,13,14] and a six-degree-of-freedom (6DOF) numerical integration technique[15]. The 6DOF integration routine incorporates the Maximum Likelihood Method (MLM) to match the theoretical trajectory to the experimentally measured trajectory. The MLM is an iterative procedure that adjusts the aerodynamic coefficients to maximize a likelihood function. The use of this likelihood function eliminates the inherent assumption in least-squares theory that the magnitude of the measurement noise must be consistent between dynamic parameters (irrespective of units). In general, the aerodynamics can be nonlinear functions of the angle of attack, Mach number, and aerodynamic roll angle.

Each model fired in the ARF was initially analyzed separately, then combined in appropriate groups for simultaneous analysis using the multiple fit capability. This provides a common set of aerodynamics that match each of the separately measured position-attitude-time profiles. The multiple fit approach provides a more complete spectrum of angular and translational motion than would be available from any one trajectory considered separately. This increases the probability that the determined coefficients define the model's aerodynamics over the entire range of test conditions.

Aerodynamic Results

For all coefficients and derivatives the reference length is the model diameter (d) and the reference area is the cross section area of the model (A). The complete set of aerodynamics for all shots is presented in Table 2.

The zero yaw axial force coefficient (C_{X_0}) versus Mach number determined from the flight data is shown in Figure 4. The shaded symbols are the results of matching multiple flight trajectories to a common set of aerodynamics. The results from the four planar fin configuration are also included. The trend-line for the planar fin data was determined via PRODAS[16], which is an aerodynamic prediction code. It was felt that since the planar fin configuration is "conventional" (i.e., easily predicted) that only a few shots were necessary and the trends predicted by PRODAS are accurate. The trend line will be moved to coincide with the measured data in this and subsequent figures.

As seen from the data, it is immediately noted that there is a substantial drag increase for the grid fin models versus the planar fin models. This trend was not unexpected and has been documented in previous papers[1-6] on grid fin aerodynamics. This increase is on the order of 100% in the subsonic regime and 67% in the supersonic regime.

Figure 5 shows the normal force coefficient derivative ($C_{N\alpha}$) versus Mach number. The grid fin results are relatively flat; indicative of grid fin behavior[4]. However, it is noted there is a decrease in the normal force coefficient derivative for that data around Mach 0.77. This is the data from shot 45 and the multiple fit data from shots 45 and 53. For the planar fin configuration, only the multiple fit result is included. Due to the low amplitude angular motion for these shots, the normal force coefficient was indeterminate based on single fits. The trend line for the planar fin data is again based upon PRODAS.

Table 2. Grid fin and planar fin aerodynamic data

Shot Number	X_{CG}/L	Air density (kg/m^3)	Mach	C_{x0}	$C_{N\alpha}$	$C_{m\alpha}$	C_{mq}	$X_{c.p.}/L$
Grid Fins								
12	0.4778	1.2012	0.574	0.695	7.630	-17.994	-	0.6252
53	0.4763	1.2030	0.744	0.771	7.620	-6.997	-	0.5337
45	0.4757	1.1974	0.771	0.812	4.860	4.424	-	0.4188
52	0.4765	1.2030	0.817	0.831	7.390	-21.594	-	0.6591
44	0.4766	1.1967	0.879	0.883	7.700	-24.094	-	0.6722
15	0.4774	1.2078	0.905	0.969	7.910	-24.134	-	0.6680
14	0.4773	1.2079	1.045	1.013	8.490	-28.067	-	0.6839
47	0.4757	1.2005	1.135	1.008	8.170	-24.158	-	0.6605
46	0.4757	1.1971	1.146	1.030	7.260	-25.920	-	0.6988
13	0.4779	1.1969	1.190	0.949	6.970	-22.438	-	0.6791
51	0.4757	1.1955	1.332	0.963	6.760	-19.647	-	0.6574
49	0.4757	1.1969	1.521	0.926	5.970	-9.409	-	0.5742
48	0.4757	1.1966	1.590	0.928	6.580	-8.868	-	0.5599
53 & 45	0.4760		0.757	0.792	4.520	0.351	-	0.4711
12, 14, & 15	0.4775		0.841	0.935	7.660	-23.024	-	0.6653
15, 44, & 52	0.4768		0.867	0.901	7.660	-22.486	-415.2	0.6603
13, 46, 47,& 14	0.4767		1.129	1.027	8.170	-25.625	-	0.6727
51, 49, & 48	0.4757		1.477	0.939	6.090	-10.799	-	0.5865
Planar Fins								
10	0.4725	1.2020	0.388	0.344	-	-36.494	-	-
11	0.4725	1.2008	0.580	0.336	9.210	-41.530	-	0.7543
9	0.4729	1.2043	0.638	0.343	10.020	-39.356	-	0.7184
8	0.4740	1.2221	0.908	0.415	14.780	-56.403	-	0.7125
9, 10, & 11	0.4726		0.535	0.341	9.860	-39.675	-663.9	0.7241

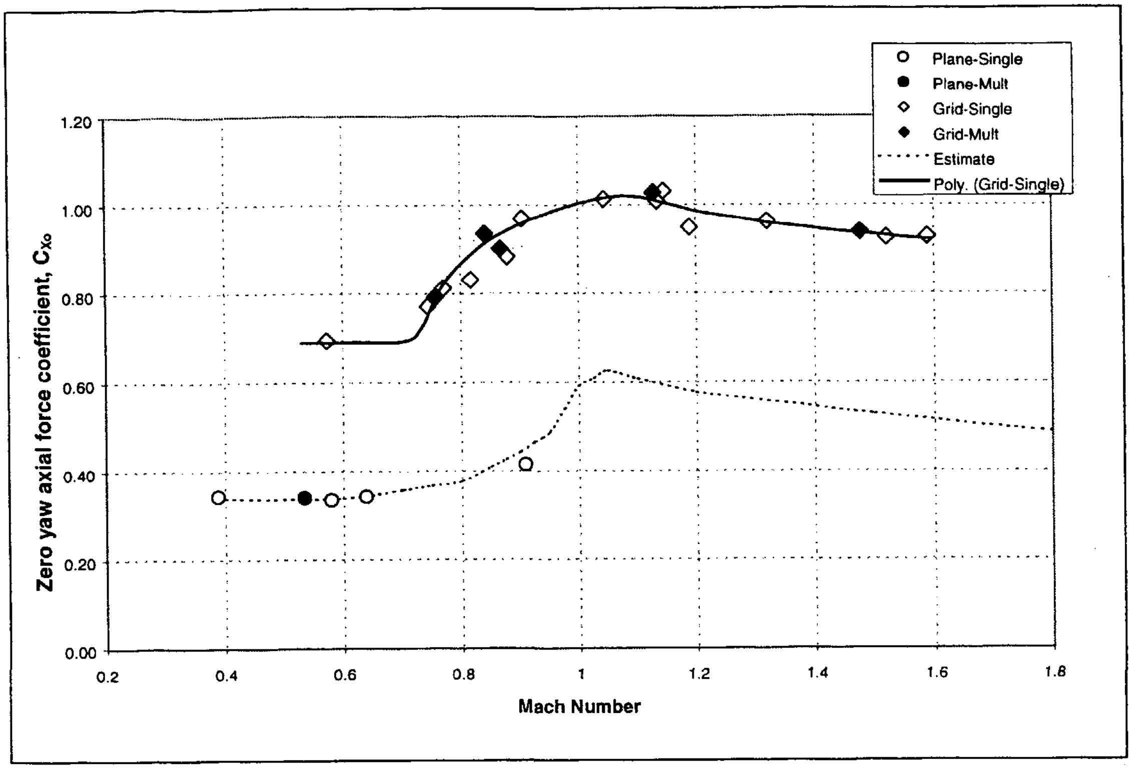

Figure 4 Zero yaw axial force coefficient versus Mach number

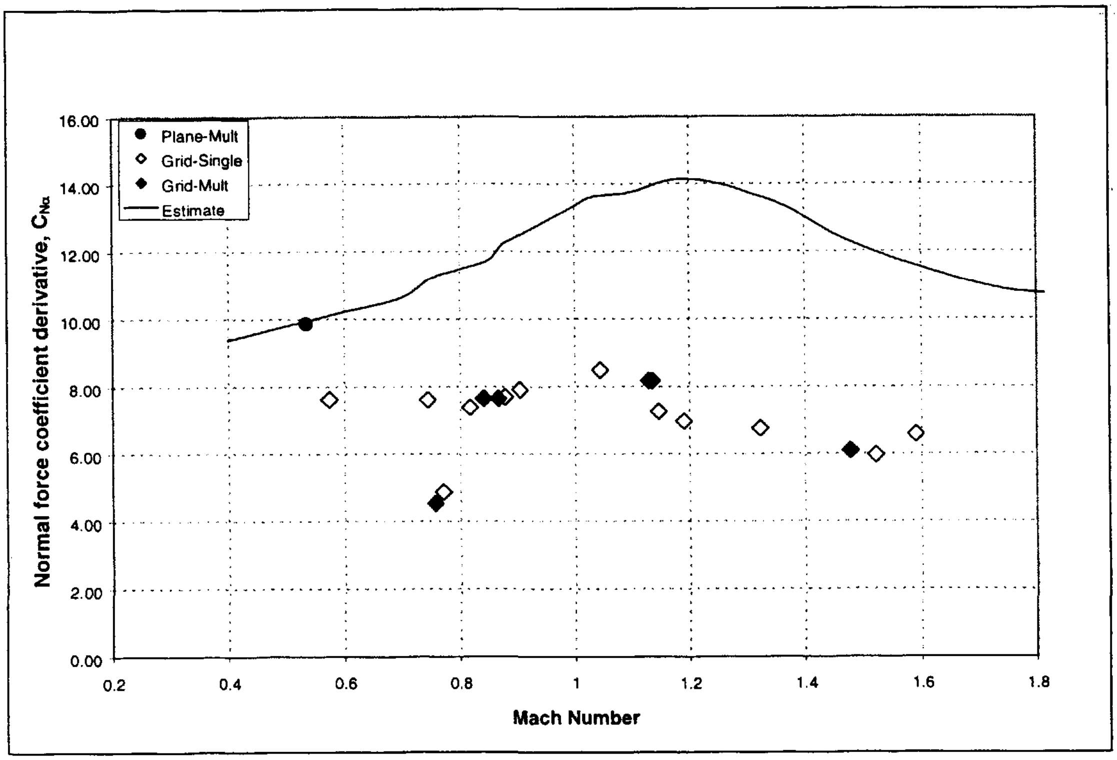

Figure 5 Normal force coefficient derivative versus Mach number

Figure 6 contains the pitching moment coefficient derivative ($C_{m\alpha}$) as a function of Mach number. Here it is seen that the trend of the grid fin data is similar to the planar fin data versus Mach number. However the grin fin data here has decreased stability for this configuration. Again note that there is a definite discontinuity in the grid fin pitch moment coefficient derivative at Mach 0.77. This data is a result of both 53 and 45. Since the angular motion within a ballistics range is measured quite accurately, this discontinuity is not attributable to measurement error. There is also noticed a loss of static stability supersonically as illustrated by the trend from Mach 1.2 to 1.6.

The angular motion in the yaw plane (ψ) as a function of downrange distance (X) is plotted in Figure 7 for shot 53 (Mach .744) and in Figure 8 for shot 45 (Mach .771). Note that for such a small change in Mach number (0.027) there is a dramatic change in the pitch frequency. Shot 53 has about one cycle of motion in 75 meters while shot 45 has one-half a pitch cycle in 100 meters. The quality of the fit of the theoretical to experimental data shown in these two figures is an indication of the confidence in the measured aerodynamics.

Pitch damping (C_{mq}) was difficult to measure in these ballistics rage tests due to the small amplitude changes during each flight in the ARF. An accurate pitch damping moment coefficient could not be determined for each shot. Trim asymmetries and roll induced side moments resulted in a very complex motion spectrum. However, based on further review of the multiple fits for groups, which had reasonable angle of attack amplitudes, the pitch moment data for the multiple fit data as shown in Table 2 are accurate to within 20%. Comparing the planar fin data to the grid fin data it is seen that there is a decrease of about 35% in pitch damping. This is reasonable to understand as the planar fins tend to act like "swash" plates within the air helping to retard pitch moment. Grid fins with their short fin chord do not resist the air as much.

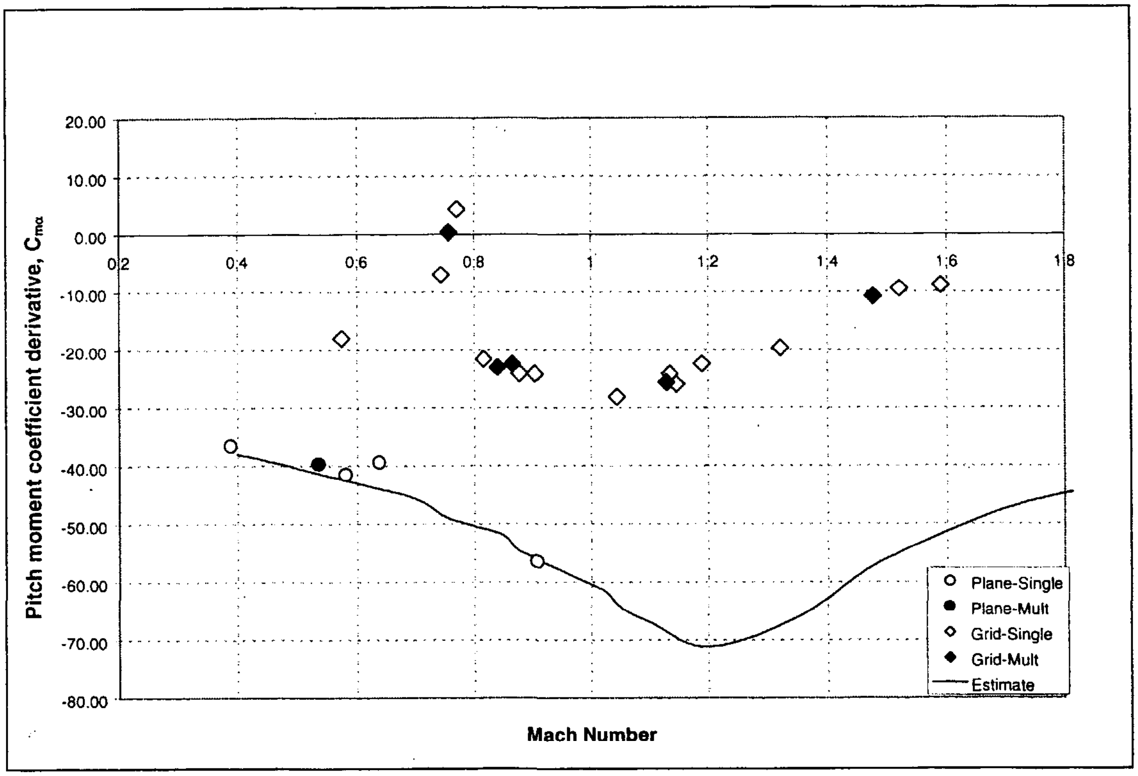

Figure 6 Pitch moment coefficient derivative versus Mach number

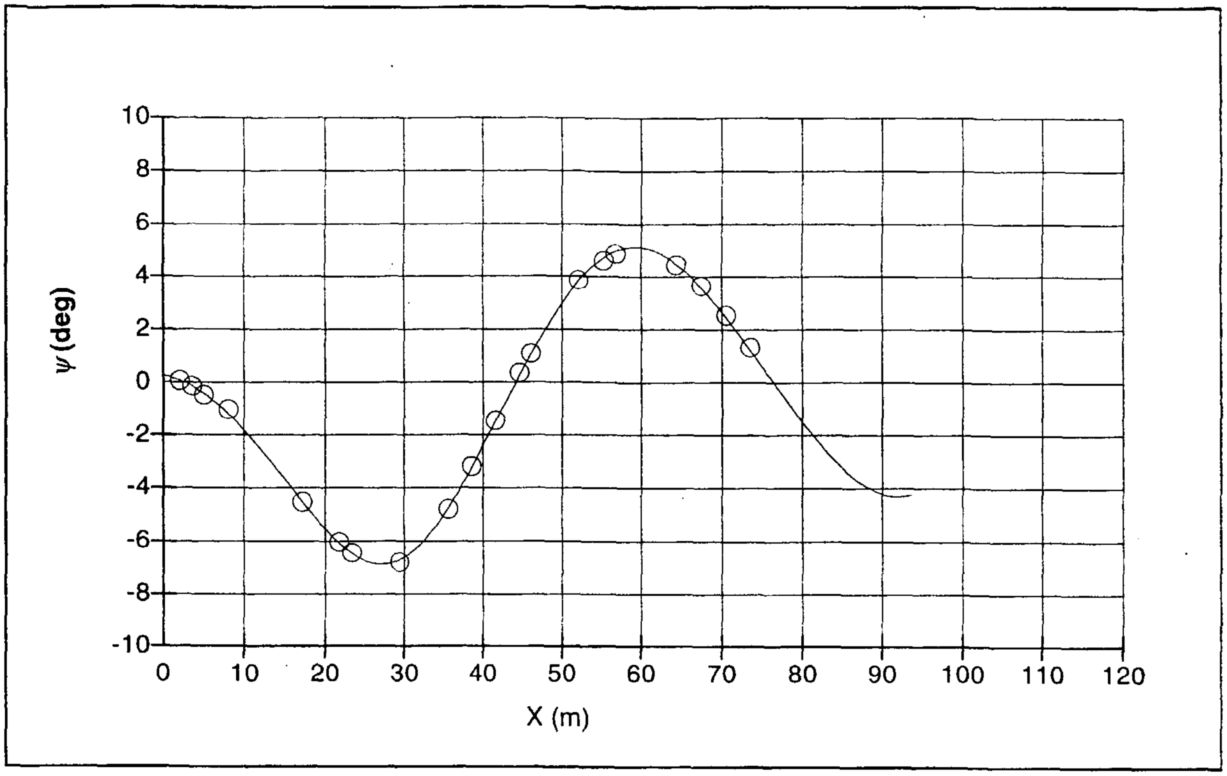

Figure 7 ψ vs. X for shot 53 (Mach 0.74)

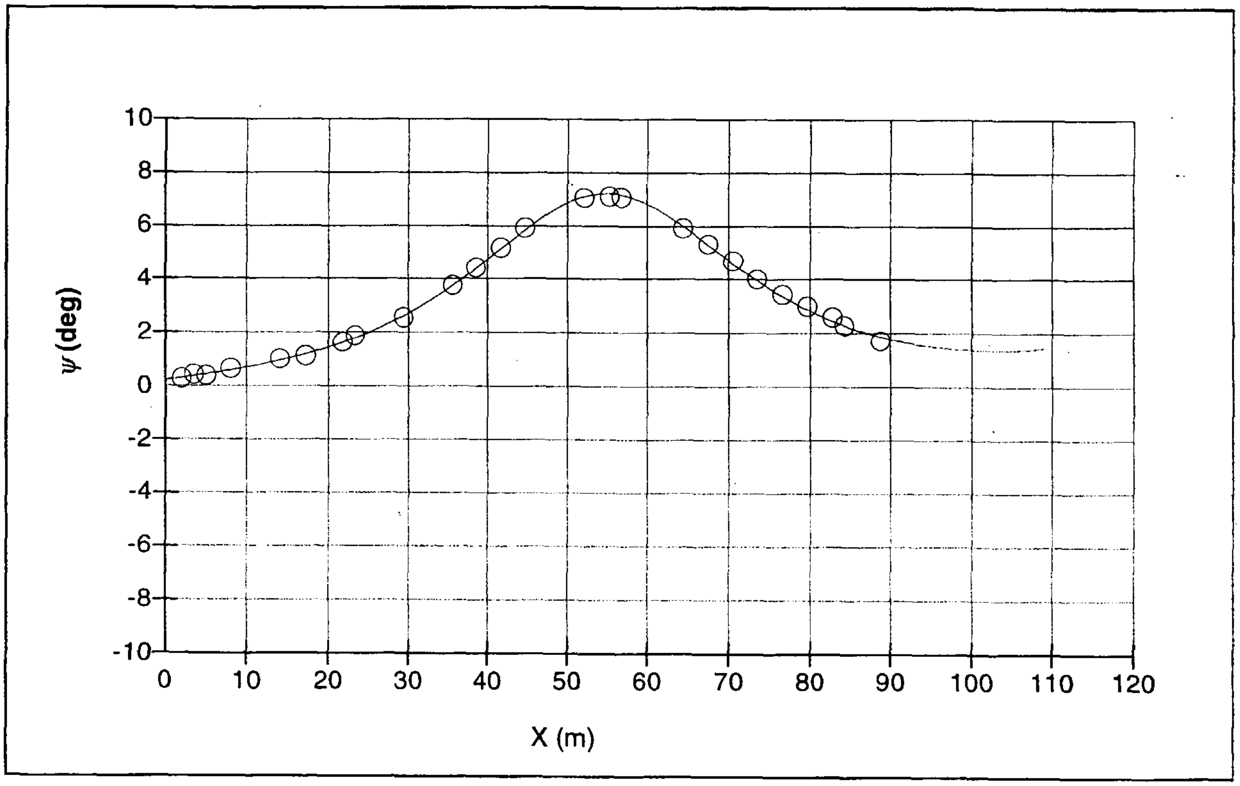

Figure 8 ψ vs. X for shot 45 (Mach 0.77)

DISCUSSION

For the current fin designs, both grid fin and planar, the aerodynamic results indicate that there is an increase in the drag for the grid fin configuration versus the planar fin. Again, this is not unexpected as grid fins are known to have increased drag versus planar fins[1-6]. However, the grid fins themselves are simple, rectangular cross-section webs. At this small scale, shaping of the individual webs was not possible. Prior results have shown that the drag on grid fins can be reduced with the proper shaping of the web cross-section[6]. Also, the grid fin web thickness is relatively thick which will also contribute to increased drag. Again, the machining process for these fins would not allow for webs much thinner than about 0.150 mm. If these fins were scaled to full scale, they would be 2-3 times as thick as necessary.

Of particular interest in the current data is the severe loss of stability at Mach 0.77. Both the normal force coefficient derivative and pitch moment coefficient derivative showed a decrease at this Mach number. The center of pressure ($X_{c.p.}$) in percent body length can be calculated for these models via (see Reference 17)

$$\frac{X_{c.p.}}{L} = \frac{X_{CG}}{L} - \frac{C_{m\alpha}}{C_{N\alpha}}\frac{d}{L}. \tag{1}$$

This has been done for this data and is given in Table 2 as well. This data is also plotted in Figure 9. Here it is seen that there is a rapid change in the center of pressure for these models at Mach 0.77. It is unknown why this occurs however. The flowfield imagery for shot 45, shown in Figure 10, shows no obvious shock waves (i.e., totally subsonic flow). However, it is not possible to know what the flowfield is like inside the grid fin cells. It is speculated that at this Mach number for this particular geometry, that the onset of supersonic flow is occurring within the fin cells thus causing the rapid change in model center of pressure. At a Mach number just above this occurrence (shot 44, Mach .879) it is seen that shocklets have formed at the tip of the fins (see Figure 11). It is interesting though, that at this Mach number (0.879) the trend in normal force coefficient derivative and pitch moment coefficient derivative is consistent with the subsonic data (i.e., the variation is smooth). Note however, that there seems to be no corresponding increase in axial force. However, the axial force data shown in Figure 4 clearly indicate that the data at Mach 0.77 is in the transonic regime. Subsonic drag coefficient data is

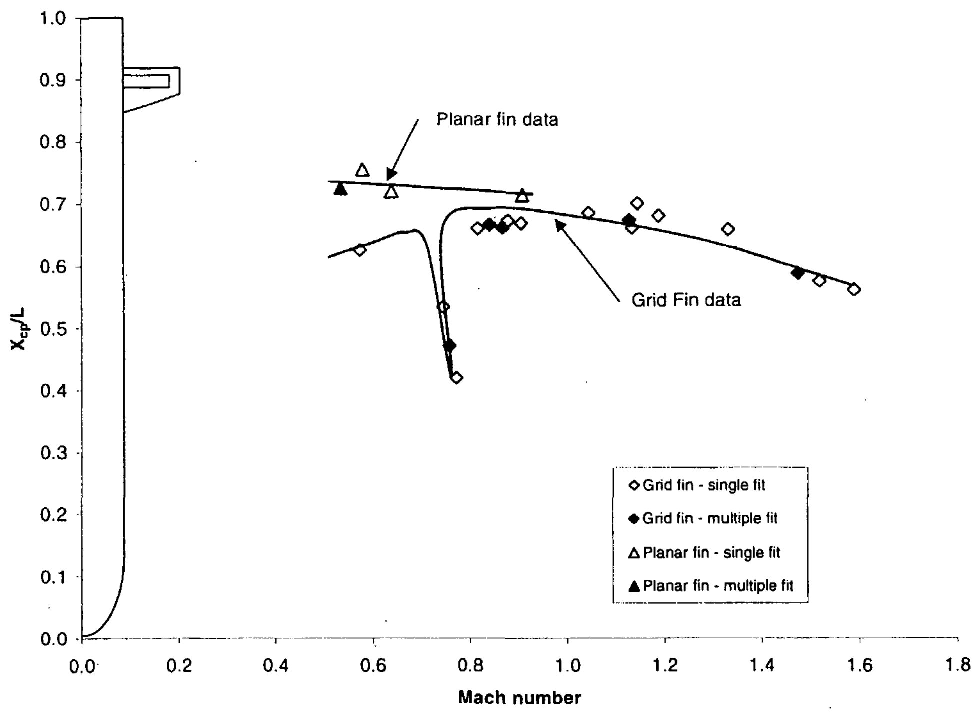

Figure 9 Center of pressure location vs. Mach number for grid fin and planar fin GTCM

constant until the onset of transonic flow. Examining the shadowgraph for this data (Figure 10) no obvious shock waves are seen. This clearly indicates that shock waves *must be present* within the grid cells. Furthermore, the fact that there are shock waves within the grid cells would explain the rapid shift in center of pressure. Since the pressure varies significantly across a shock wave. This therefore indicates that for the present data Mach 0.77 is the critical Mach number where normal shock waves have formed within the grid cells. Since this phenomena has not been seen in prior data, it is unclear if this is a result of "scaling" effects with such small grid fins or a phenomena that is present for all grid fin designs. Clearly additional testing and investigation need to be performed to understand this issue.

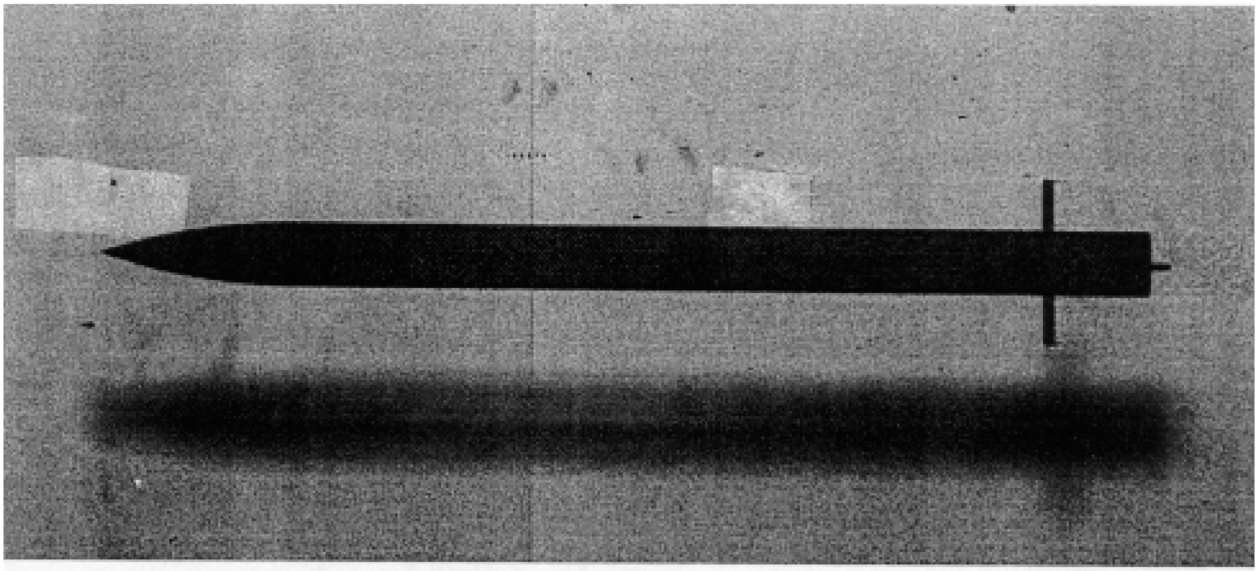

Figure 10 Shadowgraph of shot 45, station 1, Mach 0.771

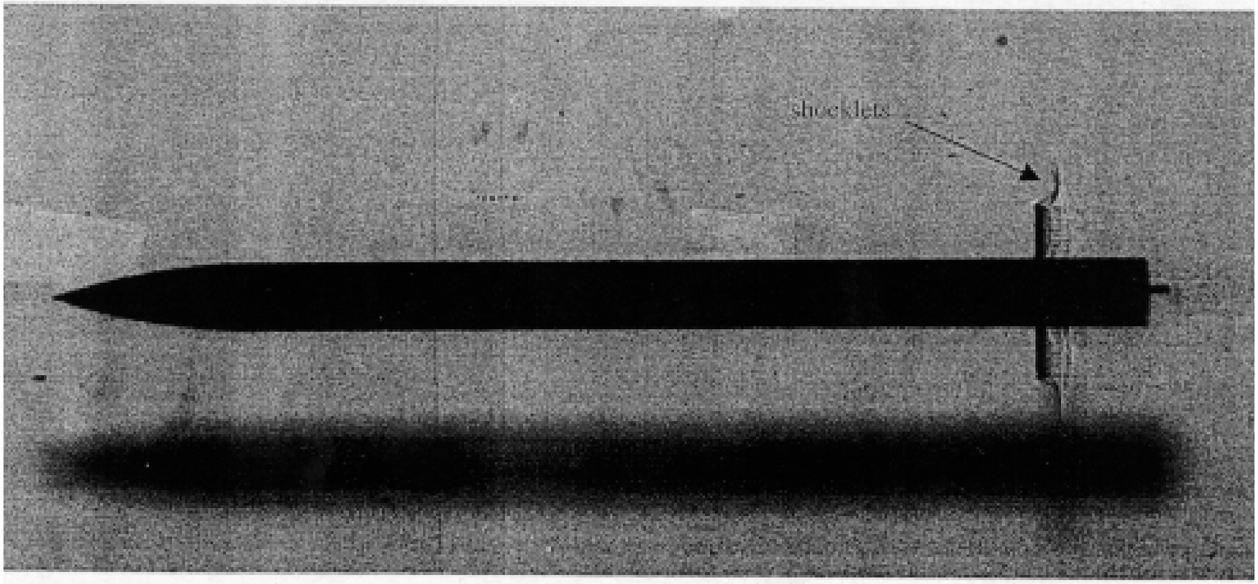

Figure 11 Shadowgraph of shot 44, station 1, Mach 0.879

CONCLUSIONS

Aerodynamic force and moment coefficients have been extracted from the measured free-flight data in the AFRL Aeroballistic Research Facility (ARF). The range of Mach numbers covered during these trials included Mach 0.39 to 1.6. A total of thirteen grid gin models were tested and evaluated at the ARF as well as four planar fin models for comparative purposes.

These experimental test results provide a good comparison of the aerodynamics of grid fins versus planar fins with respect to axial force, normal force, and pitching moment. There is a high degree of confidence in the aerodynamic data derived from these ballistics range tests. There was repeatability and a common set of aerodynamics was derived from multiple shots.

As expected the grid fin model show higher drag compared to the planar fin model. The zero yaw axial force is almost 67% - 100% higher with the grid fins. Grid fins also show lower normal force, reduced static stability, and reduced pitch damping.

The aerodynamic data also indicates that there is a critical Mach number were normal shock waves are present within the grid cells. At this particular Mach number there is seen a rapid change in the model's center of pressure. This results in severe variations of the pitch moment and normal force.

It is unclear, however, what scaling effects are present in the current data. While the data reported upon herein is believed to be highly accurate, weather or not the same results will occur for larger scaled models is unknown. Future testing in the ballistics range will be performed to address this issue.

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